After completing the preparation materials, students should be able to:

1. **Sites of action:** Map each diuretic class to its site of action in the nephron and predict the downstream effects.
2. **Mechanisms:** Explain how inhibition of specific renal transporters alters sodium handling, electrochemical gradients and downstream nephron function to produce diuresis.
3. **Effects:**
   - Correlate each diuretics class with their urinary electrolyte patterns.
   - Predict acid-base disturbances based on mechanisms.
   - Explain diuretic effects on renal hemodynamics and GFR.
4. **Compensatory responses:** Relate diuretic actions to compensatory physiological responses (e.g., SNS, RAAS activation, distal nephron reabsorption) and explain their role in diuretic resistance.
5. **Therapeutic uses:** Select the appropriate diuretic based on mechanism, patient factors, and clinical scenario.
6. **Clinical considerations:**
   - Explain how the pharmacokinetic properties influence onset, efficacy, and clinical use of the diuretic agents.
   - Predict the adverse effects and contraindications based on diuretic mechanism and patient factors.
   - Predict the impact of NSAIDs on diuretic response by describing the role of the renal prostaglandins on the actions and effects of the diuretic agents.
   - Predict drug interactions based on the mechanism of action, pharmacokinetic properties, or pharmacogenetic variabilities of the individual drugs or drug classes

Key Points Overview

- Diuretics classes are based on their sites of action along the nephron
- Diuretics increase Na^+ excretion and urine flow producing natriuresis and diuresis. They are used to adjust the volume or composition of body fluids in edematous states or other disorders, such as hypertension. Diuretics have effects on urine composition and renal hemodynamics. The extent of the effect is related to their sites of action.
- The three main diuretic classes in clinical use are **loop**, **thiazide (thiazide-type and thiazide-like),** and **potassium sparing diuretics**.
- **Carbonic anhydrase inhibitors and osmotic diuretics** are clinically important but have few therapeutic uses.
- Organic acid secretory systems located in the proximal tubule secrete diuretics from the blood into the luminal side of the tubule, where most of them act except for the mineralocorticoid receptor antagonists, which diffuse through the basolateral membrane.
- Prostaglandins (PGs) contribute to the renal physiology and to the efficacy of the diuretic agents. PGE2 and PGI2 are the major subtypes expressed in the kidney. They cause vasodilation and blunt Na^+ reabsorption and vasopressin (ADH)-mediated water transport in collecting ducts.

Key Points Diuretic Effects

- The efficacy of a diuretic is related to several factors, including its ***site*** of action, ***duration*** of action, and ***dietary salt intake***.

Diuretics can cause:

- Extracellular fluid volume contraction with compensatory responses (SNS, RAAS, ADH)
- Dehydration due to excessive water loss
- Potassium wasting or potassium sparing effect (site-dependent effects)
- Hyponatremia
- Acid-base disturbances

Stabilization with consistent diuretic therapy:

Na+ determines extracellular fluid volume (ECFV)   Steady state: Na+ intake = Na+ excretion

- Initially: Natriuresis + diuresis → ↑ urine volume and ↓ ECFV
- Compensatory mechanisms to restore Na+ balance: ↓ ECFV → activation of homeostatic mechanisms SNS, RAAS, ADH → ↑Na+ reabsorption in the nonblocked upstream segments, followed by water → sustained benefit at lower baseline ECFV

Key points CAIs

- **Mechanism:** Carbonic anhydrase inhibitors (CAIs) potently inhibit both the membrane-bound and cytoplasmic forms of carbonic anhydrase and can cause nearly complete abolition of filtered bicarbonate reabsorption in the proximal tubule along with increased secretion of sodium, potassium, and water.
- **Weak efficacy:** However, sites distal to the proximal tubule collectively have a large Na+ reabsorptive capacity. Chloride is reabsorbed in exchange for bicarbonate excretion.
- **Metabolic acidosis:** The proximal tubule is the dominant site of action of carbonic anhydrase inhibitors and the main mechanism of metabolic acidosis by CAIs. A secondary site of carbonic anhydrase inhibition in the distal nephron inhibits acid secretion and generation of new bicarbonate, which contributes to metabolic acidosis. CAIs reduce acid buffering capacity in the blood by depleting bicarbonate and, secondarily, cannot regenerate it effectively.
- **Self-limited effect:** Bicarbonate depletion reduces the amount of filtered bicarbonate, reducing drug effect. Metabolic acidosis enhances sodium reabsorption to restore sodium balance. The effects together quickly reduce CAI diuretic effect.
- **Therapeutic uses:** CAIs are used primarily in the management of open-angle glaucoma, the treatment and prophylaxis of acute mountain sickness, and metabolic alkalosis.
- Students should be aware of specific adverse effects, including potential for calcium kidney stones, and drug interactions.

Key points: Loop Diuretics

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- The loop diuretics – furosemide, bumetanide, torsemide, and ethacrynic acid – are the most potent diuretics available.
- **Mechanism:** Inhibit sodium-potassium-chloride channels (NKCC2) in the thick ascending limb (TAL) of the loop of Henle and prevent sodium reabsorption, leading to natriuresis and diuresis.
- **Efficacy:** Highly efficacious because they block up to ~25% of filtered sodium reabsorption, leading to massive fluid loss.
- **Ceiling effect:** NKCC2 saturation by the loop diuretic along with sodium reabsorption in the distal nephron limits the fraction of filtered sodium blocked by loop diuretics to ~25% of filtered Na+.
- **Therapeutic uses:** Most commonly for patients with fluid overload.
- **Drug interactions:** High potential for drug interactions by impaired entry to nephron, electrolyte disturbances or volume depletion by other drugs, changes in sodium handling by other drugs, and additive ototoxicity.

  Loop diuretics can interfere with excretion of lithium, which has a narrow therapeutic window and high potential for toxicity.
- Ethacrynic acid is a loop diuretic and the only diuretic agent that does not have a sulfonamide moiety. In the past, it may have been recommended for patients with severe allergy to sulfonamide antibiotics. Ethacrynic acid has a greater potential to cause ototoxicity than the other loop diuretics.
- Students should be aware of the numerous drug-specific adverse effects and interactions.

Class Pharmacokinetic Properties

TAKEAWAY: Pharmacokinetics is the foundation for safe prescribing, avoiding toxicity, understanding drug interactions, and personalizing treatment for your individual patients.

	Properties	**Why it matters**
A	Oral, IV, IM depending on the agent and the therapeutic use Absorption differs among the agents – some have low bioavailability and others achieve excellent plasma concentrations.	High bioavailability ensures predictable delivery to the kidney and consistent diuretic effect.
D	Highly plasma protein bound, mainly to albumin → determines free (unbound) drug concentration.	Highly protein bound diuretics are not filtered but enter the nephron via secondary active transport.
	Diuretics are secreted via OAT from blood into the proximal tubule. Sites of action are on the luminal side of renal epithelial cells. **Exception:* MR antagonists passively diffuse across basolateral membrane and bind MRs in the cytosol of the principal cells	OAT inhibitors may reduce drug concentration in nephron → may reduce diuretic efficacy.
M	Metabolism is agent specific.	Metabolism (and absorption) determines bioavailability.
E	Primarily excreted in urine as active drug.	**It makes sense:** ***The sites of action are within the nephron.***
t½	Vary by drug	Determine dosing frequency and duration of action.

Diuretics effects on electrolytes

The mechanisms are explained under the pharmacology of the specific drugs.

Diuretics modify renal handling of electrolytes:

- Hypokalemia or hyperkalemia (diuretic-specific effect)
  - K+ wasting effects: Diuretic-induced K+ (and H+) excretion
  - K+ sparing effects: Diuretic-induced K+ (and H+) reabsorption
- Hypokalemic metabolic alkalosis or hyperchloremic metabolic acidosis
- Hyponatremia
- Ca^{2+} reduced or elevated serum levels
- Hypomagnesemia
- Uric acid → hyperuricemia → ↑risk of gout in susceptible patients

Mitigate these effects by monitoring electrolyte levels and signs/symptoms

Electrolyte complications typically occur within the first 4 weeks of diuretic therapy used chronically in stable patients as the body transitions to a new steady state at a lower ECF volume.

Note: Abnormalities can occur later with dose increases, dietary Na+ intake, renal function decline, drug interactions.

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Clinical Uses of Diuretics

Edematous States	
Congestive heart failure, Liver failure Nephrotic syndrome, Acute kidney injury	Loop diuretics mobilize ECF into plasma→ kidneys→ excretion to manage volume overload
Non-Edematous States	
Hypertension	Thiazides decrease PVR.
Hypercalciuria, prevent kidney stone recurrence	Thiazides decrease Ca^{2+} excretion.
Nephrogenic diabetes insipidus	Thiazides decrease urine volume.
K^+-sparing effect	ENaC inhibitors / MRAs
Lithium toxicity	ENaC inhibitors decrease Li^+ accumulation in principal cells
Hyperaldosteronism	MRAs inhibit aldosterone
Mountain sickness / Glaucoma	CAI – Acetazolamide reduces bicarbonate availability
↑ Intracranial pressure / ↑ Intraocular pressure	Osmotic diuretic – Mannitol

PVR: peripheral vascular resistance MRA: mineralocorticoid antagonist CAI: carbonic anhydrase inhibitor

Check your knowledge

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What is the purpose of diuretic use?

What are the effects of diuretic use?

What is are compensatory effects of diuretic use?

Check your knowledge

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Where in the nephron do the potassium-wasting and potassium-sparing effects of diuretics occur?

What is the key determinant of potassium secretion in the distal nephron?

A 68-year-old man with chronic heart failure is treated with furosemide for worsening leg edema. After starting ibuprofen for back pain, his urine output decreases and his weight increases. **What mechanism best explains why ibuprofen reduced the effectiveness of his diuretic?**

CA inhibitors cause mild hyperchloremic metabolic acidosis and self-limited diuretic effect.

Metabolic acidosis:

CA inhibition leads to massive bicarbonate loss in urine.

- Continued bicarbonate loss reduces plasma bicarbonate → ↓ arterial pH

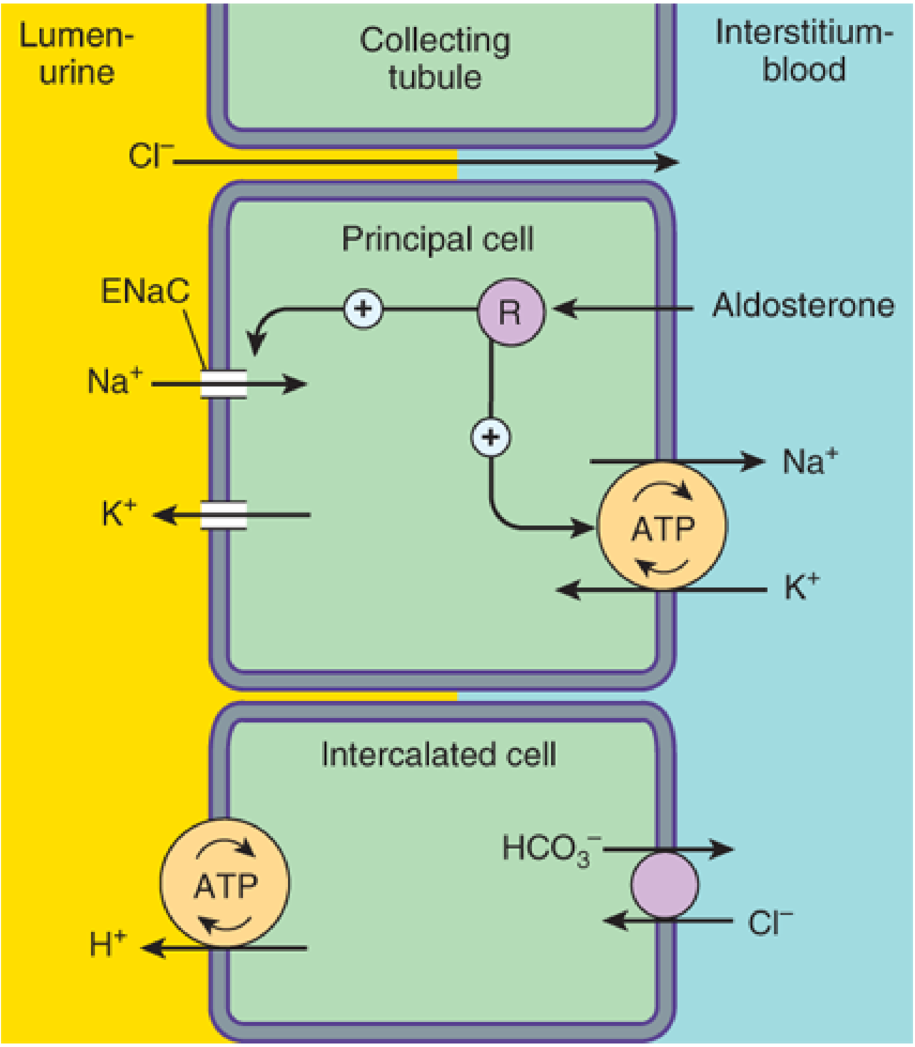
Chloride is reabsorbed to maintain electroneutrality.

hyperchloremic metabolic acidosis

Self-limited diuretic effect (2-4 days of continuous therapy)**:**

1. **Low filtered HCO3– concentration:** Large bicarbonate loss in urine causes plasma HCO3– levels to fall dramatically.
   - Lower plasma HCO3– → lower filtered HCO3– → ↑Na+ reabsorption in the proximal tubule → ↓ Na+ excretion → ↓natriuresis and diuresis
   - Cl– is reabsorbed along with Na+.
2. ↓ **ECF volume + Metabolic acidosis:** RAAS / aldosterone / ADH stimulates ↑ENaC expression + ↑AQP2 insertion in collecting duct→ ↑Na+ and water reabsorption

reduced diuretic efficacy



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition

Slide adapted by LG

Potassium-wasting:

↑ Na+ reabsorption in the collecting duct → lumen becomes more negative → ↑ K+ secretion

Changes in Urinary Electrolyte Patterns and Body pH in Response to CAIs				
	Urinary Electrolytes:			
Carbonic anhydrase inhibitors	NaCl	HCO3	K+	Body pH
	+	+++	+	↓

+, increase; –, decrease; 0, no change; ↓, acidosis; ↑, alkalosis
Adapted from Katzung's Basic & Clinical Pharmacology Table 15-2

K+: potassium wasting effect

Body pH: mild metabolic acidosis

RECAP: Effects of CAIs
↑ Na^+ excretion with water→ mild natriuresis / diuresis
↑ HCO_3^- excretion → ↑urine pH / ↓ total body HCO_3^- concentration → mild acidosis
↑ Cl^- reabsorption to maintain electroneutrality → ↑ serum Cl^- concentration → hyperchloremia
Aldosterone-mediated ↑ENaC expression and Na+ reabsorption →↑ K^+ excretion → ↓ serum K^+ concentration → mild hypokalemia
CAIs reduce Ca^{2+} tubular reabsorption / increase Ca^{2+} excretion → may predispose patients to kidney stones. CAIs have little to no effect on Mg^{2+} excretion.
CAIs increase solute delivery to TAL / macula densa → strong activation of TGF mechanism → afferent arteriolar vasoconstriction → ↓ GFR (sustained CAI use over time leads to TGF adaptation that blunts effect)

Clinical Applications of CA Inhibitors

- **Acute mountain sickness, prevention and treatment:**

  Mild metabolic acidosis stimulates chemoreceptors and respiratory center → increases minute ventilation → increase CO2 exhalation → ***sustained hyperventilation and improved oxygenation*** (PaO2)

  → high altitude symptoms diminish

  Mountain sickness results from travel to a high altitude too quickly to adjust to reduced oxygen levels. Symptoms include shortness of breath, headache, fatigue, dizziness, nausea, and insomnia. Pulmonary edema and cerebral edema can occur in severe cases.
- **Glaucoma, open-angle:**

  Reducing bicarbonate production in the ciliary body reduces aqueous humor secretion, which decreases intraocular pressure for prevention of optic nerve damage.
- **Metabolic alkalosis:**

  Rapid correction, especially loop diuretic-induced, which is caused by volume contraction, increased bicarbonate reabsorption, and aldosterone-stimulated H^+ secretion.
- **Edema:** ***Low efficacy*** Long-term treatment is complicated by metabolic acidosis.

→ **CAIs remain highly effective in the treatment of mountain sickness** because HCO_3^- reabsorption is prevented and buffering capacity is depleted, and in glaucoma as bicarbonate production is inhibited.

- Only the natriuretic / diuretic effect of CAIs is self-limiting because it depends on filtered bicarbonate, which becomes depleted. (Metabolic acidosis is not self-limited.)

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Check your knowledge of carbonic anhydrase inhibitors

CAIs significantly increase the excretion of what electrolytes?

What is the effect of CAIs on the pH of blood?

How is the CAI’s affect on blood pH used to therapeutic advantage?

Check your knowledge of carbonic anhydrase inhibitors

Why do CAIs cause only mild diuresis?

Why is the diuretic effect of CAIs self-limited?

Why is the action of CAIs not self-limited in the treatment of glaucoma and mountain sickness?

A 28-year-old first-time mountain climber starts acetazolamide for prevention of acute mountain sickness before ascending to high altitude. Two days later, he reports a tingling “pins and needles” sensation in his fingers and around his lips but otherwise feels well.

1) What is the side effect likely caused by the medication?

2) What is the mechanistic rationale for the medication for this purpose?

Loop Diuretics: Pharmacokinetic Properties

TAKEAWAYS: What to use for refractory edema and why. PK properties are typical of the class (slide 15).

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Furosemide	Torsemide	Ethacrynic acid
Oral, S/L, IV infusion, subQ infusion Oral, S/L absorption ~50-60% variable absorption **Why it matters:** In ***heart failure*** patients, GI edema, reduced mucosal blood flow, and slowed gastric emptying can significantly reduce absorption, therefore, decrease effectiveness.	Oral only ~100% absorption and bioavailability **Why it matters:** In patients with ***refractory edema*** unresponsive to furosemide, it may be reasonable to switch to torsemide, which has more predictable absorption and a longer half-life than furosemide or bumetanide.	**Takeaways:** (boards and practice relevant) • Lacks sulfonamide moiety. It was once thought that ethacrynic acid may be preferred in patients with severe sulfonamide allergy. • Ethacrynic is NOT more effective than the other loop diuretics. • It does NOT have pharmacokinetic advantages compared to the others. • Ototoxicity risk is HIGHER, especially in combination with aminoglycosides. • Very expensive • Availability may be limited.
90-99% protein bound OAT1, OAT3 secretion Metabolized in kidney Excreted in urine t½ 1.5-2h Diuresis duration oral ~6 h, IV ~2 h	>99% protein bound OAT secretion Hepatic metabolism Excreted in urine t½ 3-5 h Diuresis duration oral ~12 h	

S/L: sublingual

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Physiological responses to loop diuretics (NKCC2 inhibition)

Renal Hemodynamics	Impaired Urine Concentration / Dilution	Diuretic Resistance
Acute effect: Loop diuretics stimulate intrarenal PGE2/PGI2 secretion → RBF increases briefly and GFR is preserved **Later effect:** Volume contraction → RAAS activation → mild decline in RBF and GFR	NKCC2 blockade causes: • collapse of the medullary gradient → collecting duct cannot reabsorb water effectively even with ADH present and • TAL does not dilute tubular fluid so urine diluting capacity is lost	Edema refractory to typically effective doses of loop diuretic occur due to • compensatory tubular adaptations, • renal hemodynamics, and • PK mechanisms that reduce sodium and water excretion.

Recall: PGE_2 / PGI_2 participate in local vasodilation, renin release, and salt and water transport → PGE2 and PGI2 **participate in the actions of loop diuretics**.

➢ ***NSAIDs reduce the efficacy of loop (and other) diuretics.***

Edema refractory to typically effective doses of loop diuretic

Diuretic resistance is most prominent with loop diuretics. Only loop diuretics trigger nephron remodeling and compensatory sodium retention sufficient to produce clinically significant diuretic resistance.

Mechanisms: compensatory tubular adaptations, renal hemodynamics, and PK mechanisms

Illustrations are on next slide

Primary mechanism with chronic use:

- Distal nephron adaptation by hypertrophy of distal segments and upregulation of downstream Na+ transport: NCC, ENaC, and activity of Na+/K+ ATPase → ↑Na+ reabsorption capacity

Amplification:

- Volume depletion → Neurohumoral activation of compensatory mechanisms: RAAS activity, aldosterone secretion, and ADH actions
- Macula densa suppression → stimulates renin release

Contributing factors:

- Decreased absorption (gut edema heart failure patients) → reduced drug delivery to sites of action
- Impaired proximal tubular secretion (OATs)
- Higher sodium intake
- Drug interactions, especially NSAIDs and probenecid (OAT inhibitor)

Diuretic Resistance

"diuretic braking phenomenon"

Diuretic Resistance: when kidneys fail to increase sodium and fluid excretion despite maximal diuretic doses

Mechanisms: pharmacokinetic, renal hemodynamics, and compensatory tubular adaptations

PK: ↓ Absorption reduces drug delivery to nephron / short t½ → limited duration

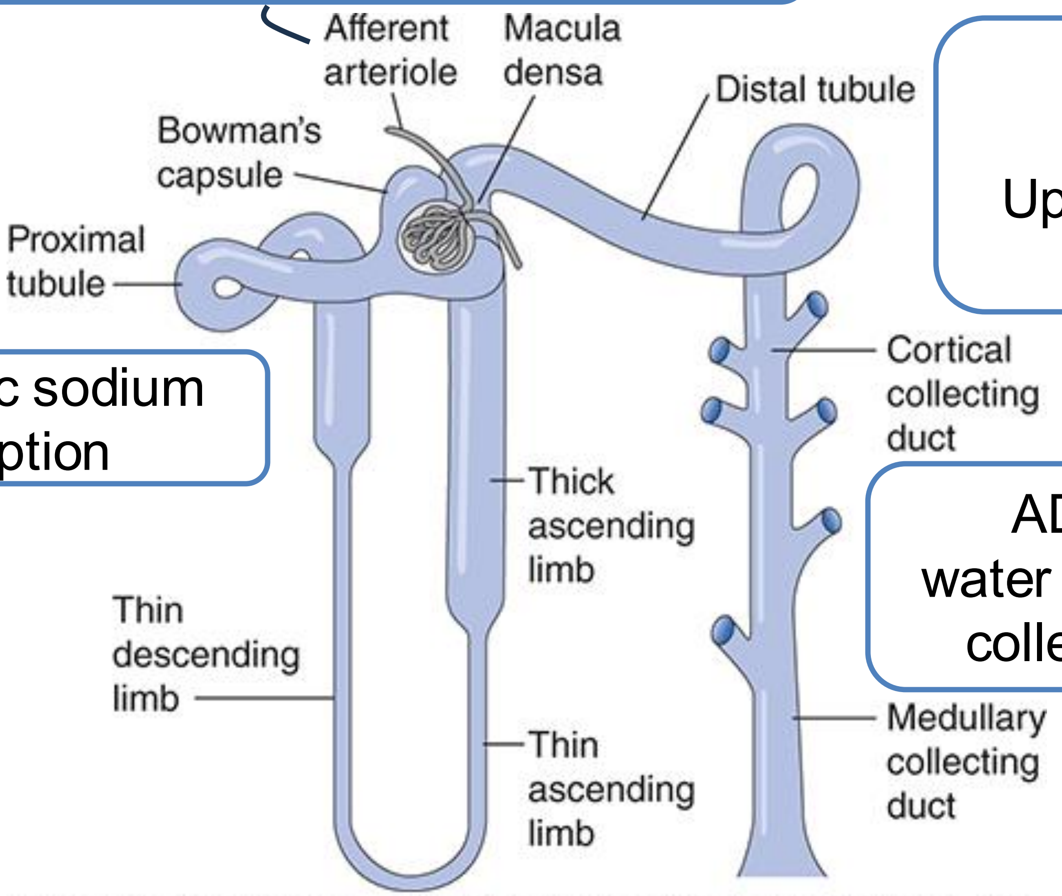
↓ renal perfusion pressure
→ ↓ drug delivery to site of action

Post-diuretic sodium reabsorption

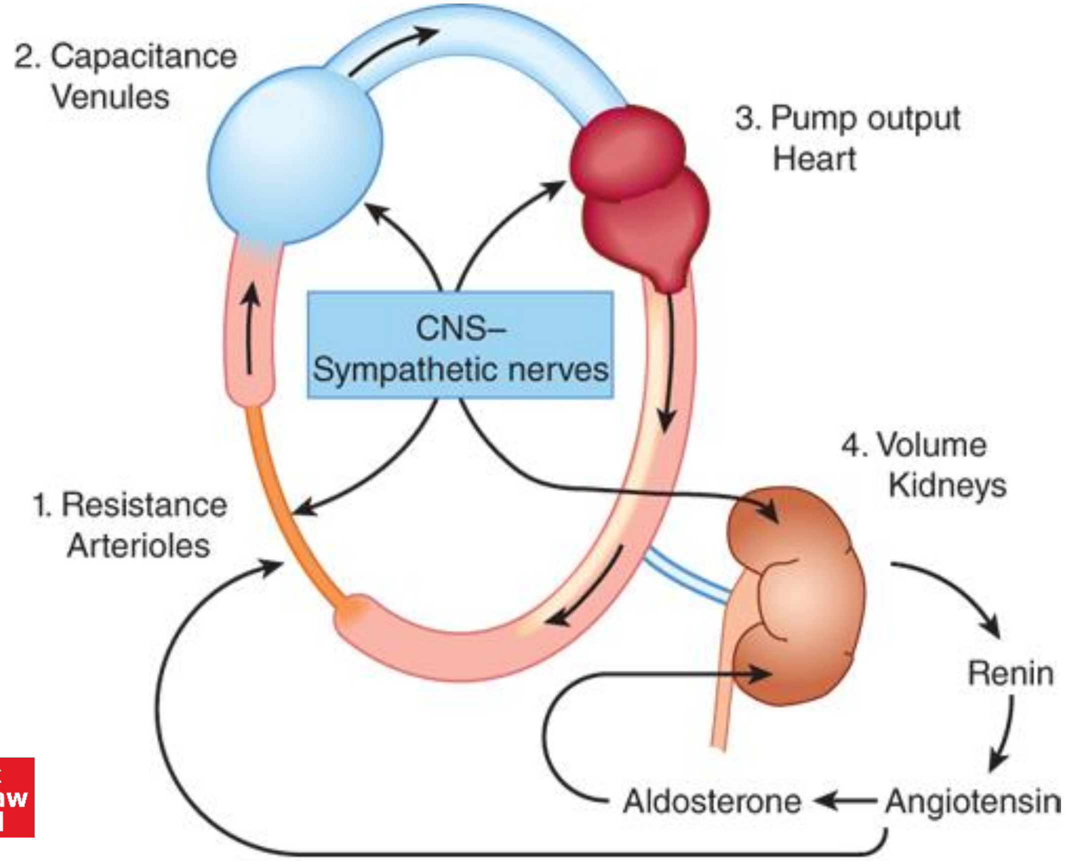
Distal nephron hypertrophy
Upregulation of Na+ transporters

ADH-induced water reabsorption in collecting tubules

↓ECF volume → RAAS / aldosterone
→ Na+/water reabsorption
and
ADH actions → water retention
and other actions



Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*



Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition

ECF: extracellular fluid
SNS: sympathetic nervous system
RAAS: renin-angiotensin-aldosterone system

Changes in Urinary Electrolyte Patterns and Body pH in Response to CA Inhibitors and Loop Diuretics

Diuretic Class	Urinary Electrolytes NaCl	HCO_3^-	K^+	Body pH
Carbonic anhydrase inhibitors	+	+++	+	↓
Loop diuretics	++++	0	+++	↑

+, increase; –, decrease; 0, no change; ↓, acidosis; ↑, alkalosis

Adapted from Katzung's Basic & Clinical Pharmacology Table 15-2

RECAP: Effects of Loop Diuretics
↑ Na^+ excretion (up to 25% of filtered sodium) and water follows → profound natriuresis / diuresis
K+-wasting effect, ECF volume contraction and RAAS activation → hypokalemic metabolic alkalosis
Loop diuretics cause a marked increase in excretion of Ca^{2+} and Mg^{2+} due to elimination of transepithelial potential. Calcium loss is usually mild. Magnesium loss may be clinically significant.
Aldosterone-mediated ↑ENaC expression and Na+ reabsorption →↑ K^+ excretion → ↓ serum K^+ concentration → may require potassium supplementation or potassium-sparing strategy.
Loop diuretics impair the kidney's ability to concentrate urine and excrete dilute urine.
Loop diuretics are associated with diuretic resistance.

Loop Diuretics: Clinical Applications

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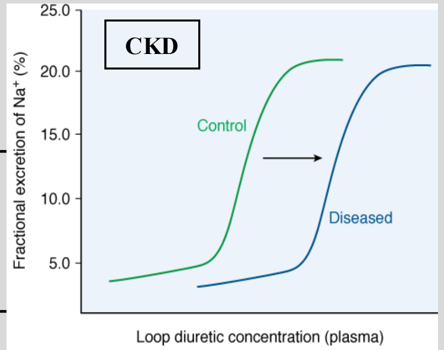
Clinical use	Effects and comments
Acute pulmonary edema	Natriuresis + venodilation → ↓ LV pressure and pulmonary edema
Chronic CHF	↓ Effective circulating volume minimizes venous / pulmonary congestion
Nephrotic syndrome	Relief of edema due to salt / water retention
Ascites of liver cirrhosis	Spironolactone with furosemide (oral) and salt restriction
Acute kidney injury	To relieve signs/symptoms of fluid overload Diuretic is not a replacement for dialysis.
Chronic kidney disease	Higher doses often needed (fewer nephrons, OAT competition (uremia), tubular adaptations)
Hypertension	Thiazides are a guideline-directed first-line antihypertensive Loop diuretics → risk of volume/electrolyte depletion / short duration BP lowering effect
Chemotherapy toxicity	Forced diuresis with loop diuretic and saline infusion to prevent chemo-induced nephrotoxicity *Potential complications of forced diuresis: rapid shifts in fluid/electrolyte balance can paradoxically lead to fluid overload, thus, pulmonary edema in addition to electrolyte disturbances, especially in patients with reduce heart, kidney, or lung function.*



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*: www.accessmedicine.com

Loop Diuretics: Adverse Effects

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Most are due to abnormalities of fluid and electrolyte balance.

- Hypomagnesemia (common, dose-dependent)
- Hyponatremia (milder and less frequent than with thiazides)
- Hypokalemia and metabolic alkalosis

Mechanism of hypokalemic metabolic alkalosis: Loop diuretics cause metabolic alkalosis through potassium loss and increased H+ secretion (via aldosterone), volume contraction, and RAAS activation that promotes proximal tubule bicarbonate reabsorption. Chloride loss with bicarbonate generation in collecting tubule maintains metabolic alkalosis.

Other toxicities:

- Ototoxicity (inhibition of NKCC1 in the inner ear disrupt endolymph K+ balance essential for normal hearing)
- Hyperuricemia (milder and less frequent than with thiazides)
- Hypersensitivity reactions

Pregnancy: Caution – risk of decreasing placental perfusion

Diuretics reduce maternal plasma volume and cardiac output, which can lead to decreased uteroplacental blood flow and cause electrolyte imbalances.

- Diuretics should be used cautiously if benefit to the mother outweighs risk to the fetus.
- Monitor fetal growth and maternal volume status if diuretics are used.

Drug Interactions involving Loop Diuretics

Concomitant drugs	Effect
Thiazide diuretics	Diuretic synergism can lead to profound diuresis (usually desirable)
Antihypertensive agents	Enhanced hypotensive effect (usually desirable)
ACE inhibitors / ARBs	Enhanced nephrotoxicity Volume depletion in the presence of inhibitors of angiotensin II-mediated constriction of the efferent arterioles can induce acute renal injury.
Lithium (Li^+)	LI^+ is absorbed with Na^+ mainly in the proximal tubule. Enhanced Na^+ excretion → ↑Li^+ absorption → lithium toxicity (Loop diuretics have a smaller effect on lithium levels than thiazides.)
Digitalis glycosides	Hypokalemia →↑ digitalis toxicity → arrhythmias
Quinidine and other QT interval prolonging drugs	Diuretic-induced hypokalemia increases the risk of torsades de pointes

Drug Interactions involving Loop Diuretics

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Concomitant drugs	Effect and Mechanism
Antidiabetic agents	Impaired glucose control May involve hypokalemia wth impaired insulin release and glucose uptake → hyperglycemia
Probenecid	Competition for proximal tubule transporters → blunted diuretic response and ↑ loop diuretic's adverse effects
NSAIDs	NSAIDs ↑ Na^+ and water reabsorption → blunted diuretic response and can worsen metabolic alkalosis (loss of PG-mediated afferent dilation is perceived as low perfusion → RAAS activation →→→ ↑K+, H+ secretion / bicarbonate reabsorption) Prostaglandins participate in the renal actions of loop diuretics
Glucocorticoids	Increased risk of hypokalemia Glucocorticoids promote Na^+ retention and K^+ loss
Amphotericin B	Increased risk of hypokalemia Amphotericin-induced renal injury induces potassium loss
Aminoglycosides	Synergism of nephrotoxicity and ototoxicity
Cisplatin	Synergism of nephrotoxicity and ototoxicity

Check your knowledge

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How does blocking the NKCC2 lead to high-efficacy diuresis?

What are the mechanisms of the loop diuretics’ effect on urinary electrolytes?

Loop diuretics can cause metabolic alkalosis. What is the mechanism?

Check your knowledge

What are the contributors to edema refractory to loop diuretic therapy?

Which patients are susceptible to loop diuretic-refractory edema?

What is the mechanism by which loop diuretics impair the kidney’s ability to produce dilute urine during high intake of water and concentrated urine during hydropenia?

A 75-year-old male with hypertension and heart failure is taking lisinopril (ACE inhibitor) and furosemide. After self-treating chronic knee pain with naproxen, the patient develops dizziness and a sharp rise in serum creatinine.

Explain the mechanism of the drug-drug interaction leading to the adverse drug event.

- The goal of diuretic therapy is to restore and maintain euvolemia by altering the kidney's regulation of sodium and water. Reducing extracellular fluid volume and blood volume by stimulating natriuresis and diuresis relieves fluid overload and lowers blood pressure.
- Carbonic anhydrase inhibitors (CAIs) (*eg,* acetazolamide) prevent bicarbonate and sodium reabsorption in the proximal tubule. Although inhibiting the reabsorption of 65% to 70% of filtered sodium, they are mild diuretics because most of the sodium is reabsorbed in the more distal nephron segments.
- CAIs increase urinary excretion sodium, bicarbonate, potassium, and calcium. CAIs increase urine pH as bicarbonate is excreted.
- CAIs mild hyperchloremic metabolic acidosis by abolishing bicarbonate reabsorption and mild hypokalemia (potassium-wasting effect).
- CAIs are used for prophylaxis and treatment of mountain sickness by increasing ventilation, alkalosis by excretion of bicarbonate in the urine, and open-angle glaucoma by inhibiting aqueous humor secretion, which reduces intraocular pressure.

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- Loop diuretics (furosemide, bumetanide, torsemide) inhibit the NKCC2 in the TAL preventing the reabsorption of ~25% of filtered sodium, which results in profound and rapid natriuresis, thus profound diuresis.
- Loop diuretics increase urinary excretion of sodium, potassium, calcium, and magnesium.
- Loop diuretics potassium-wasting effect can lead to hypokalemia.
- Loop diuretics cause metabolic alkalosis through hypokalemia, chloride depletion, and volume contraction with RAAS activation, which, together with increased H^+ loss, stimulate ammoniagenesis and impair bicarbonate excretion, which results in increased generation and retention of new bicarbonate.
- Loop diuretics are used in the treatment of edema, heart failure, and hypercalcemia.
- Loop diuretics are effective in lowering blood pressure, especially in heart failure, kidney disease, and cirrhosis. They are not first-line for management of primary hypertension because of short duration of action of adverse effects.

Check your knowledge of general principles

What is the purpose of diuretic use?

- To decrease extracellular fluid volume by increasing sodium and water excretion by the kidneys, with the goal of restoring and maintaining euvolemia.

What are the effects of diuretic use?

- Edema reduction: Diuretics induce natriuresis / diuresis, which promotes movement of fluid from the interstitial space into the intravascular compartment, where it can be excreted by the kidneys promotes renal excretion, thereby restoring normal effective circulating volume (euvolemia).
- Blood pressure lowering effect (thiazide diuretics): Diuretics reduce effective circulating volume by increasing renal sodium and water excretion, which lowers venous return and cardiac output initially. Over time, peripheral vascular resistance decreases → sustained reduction in blood pressure.
- Diuretics affect electrolyte balance – sodium, potassium, calcium, magnesium, bicarbonate. The

What is are compensatory effects of diuretic use?

- Volume depletion activates the RAAS cascade, resulting in aldosterone-enhanced sodium and water reabsorption, counteracting the diuretic effect.
- Volume contraction → reduced plasma volume, which educed plasma volume may stimulate secretion of vasopressin, which increases free water reabsorption in the collecting ducts.

Check your knowledge

Where in the nephron do the potassium-wasting and potassium-sparing effects of diuretics occur?

- K+ secretion is strongly regulated by aldosterone in principal cells in the late distal tubule and collecting duct.

What is the key determinant of potassium secretion in the distal nephron?

- Increasing Na+ reabsorption by ENaC creates a lumen-negative transepithelial potential that favors K+ secretion via ROMK.

A 68-year-old man with chronic heart failure is treated with furosemide for worsening leg edema. After starting ibuprofen for back pain, his urine output decreases and his weight increases. **What mechanism best explains why ibuprofen reduced the effectiveness of his diuretic?**

- NSAIDs block renal PGE2 / PGI2 synthesis → afferent vasoconstriction → ↓RBF and ↓GFR

By blocking COX-1, COX-2 and PG synthesis, NSAIDs prevent prostaglandin-mediated afferent vasodilation → reduces renal blood flow, GFR, and diuretic delivery, leading to blunted diuretic effect.

- NSAIDs inhibit OATs, reducing delivery of the drug into proximal tubular fluid.
- NSAIDs cause sodium and water retention, reducing diuretic efficacy.

Check your knowledge of carbonic anhydrase inhibitors

CAIs significantly increase the excretion of what electrolytes?

- CAIs increase urinary concentrations of **bicarbonate, sodium, potassium, and calcium.**
- Carbonic anhydrase reversibly catalyzes the forward and reverse reaction of $CO_2 + H_2O \rightleftharpoons H_2CO_3^- \rightleftharpoons H^+ + HCO_3^-$. CA is essential for reabsorption of 80% to 90% filtered bicarbonate and generation of H^+ that drives Na+ reabsorption.

What is the effect of CAIs on the pH of blood?

- Mild hyperchloremic metabolic acidosis
- Increased excretion of bicarbonate in the urine reduces serum bicarbonate, which is a key buffer in maintaining pH of the blood, leading to mild metabolic acidosis.
- Reabsorption of filtered chloride increases to maintain electroneutrality, which leads to a relative increase in serum chloride. Thus, CAIs cause hyperchloremic metabolic acidosis.

How is the CAI’s affect on blood pH used to therapeutic advantage?

- Prevention and treatment of acute mountain sickness (acute altitude sickness).
- Mild metabolic acidosis increases stimulates ventilation, increasing CO_2 exhalation, which sustains hyperventilation and enhances oxygenation. High altitude symptoms diminish.

Check your knowledge of carbonic anhydrase inhibitors

Why do CAIs cause only mild diuresis?

- CAIs inhibit reabsorption of bicarbonate in the proximal tubule along with Na+.
- The nephron distal to the proximal tubule has a large Na+ reabsorption capacity, so only a small percentage of filtered Na+ is excreted, leading to mild natriuresis and diuresis.

Why is the diuretic effect of CAIs self-limited?

- CA inhibition leads to bicarbonate loss in urine → metabolic acidosis → over time plasma bicarbonate decreases → filtered bicarbonate load falls and, with it, Na+ excretion is reduced.
- ECF volume contraction plus metabolic acidosis stimulates aldosterone secretion → ENaC upregulation → increased Na+ reabsorption further limiting the natriuretic / diuretic effect.
- **Thus, diuretic efficacy is reduced.**

Why is the action of CAIs not self-limited in the treatment of glaucoma and mountain sickness?

- Acute mountain sickness treatment is due to induction of metabolic acidosis.

Metabolic acidosis is not a self-limited effect because HCO_3^- reabsorption continues to be prevented and buffering capacity of the blood is depleted.

- Glaucoma treatment goal is reduction of intraocular pressure by inhibiting bicarbonate formation, which reduces aqueous humor production.

A 28-year-old first-time mountain climber starts acetazolamide for prevention of acute mountain sickness before ascending to high altitude. Two days later, he reports a tingling “pins and needles” sensation in his fingers and around his lips but otherwise feels well.

1) What is the side effect likely caused by the medication?

2) What is the mechanistic rationale for the medication for this purpose?

- 1) paresthesia
- 2) Mechanism:

With rapid ascent to high elevation (~8200 ft) the body does not have time to adjust to the lower oxygen pressure, which can cause headache, nausea/vomiting, fatigue, dizziness, sleep disturbance.

The body responds to hypoxia sensed by peripheral chemoreceptors, resulting in hyperventilation. CO_2 exhalation leads to hypocapnia, which leads to respiratory alkalosis, which blunts peripheral chemoreceptors and inhibits the respiratory center, limiting further hyperventilation.

- Inhibiting carbonic anhydrase causes mild metabolic acidosis, which stimulates peripheral chemoreceptors and respiratory center → increases minute ventilation → ***sustained hyperventilation and improved oxygenation*** (PaO_2)

High altitude symptoms diminish.

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Check your knowledge of loop diuretics

How does blocking the NKCC2 lead to high-efficacy diuresis?

- ~25% of Na^+ reabsorption occurs in the TAL via the NKCC2. Blocking this transporter results in profound and rapid excretion the sodium load, which results in profound diuresis.

What are the mechanisms of the loop diuretics' effect on urinary electrolytes?

- Inhibition of NKCC2 by loop diuretics prevents reabsorption of sodium, chloride and potassium and eliminates the positive lumen potential preventing paracellular transport of Ca^{2+} and Mg^{2+}. Therefore, sodium, chloride, potassium, calcium and magnesium are excreted in urine.

Loop diuretics can cause metabolic alkalosis. What is the mechanism?

- Loop diuretics cause metabolic alkalosis through potassium loss and increased H+ secretion (regulated by aldosterone), volume contraction, and RAAS activation that promotes proximal tubule bicarbonate reabsorption. Hypochloremia impairs bicarbonate excretion and increased H+ secretion generates and retains bicarbonate, maintaining loop diuretic-induced alkalosis.

Check more of your knowledge about loop diuretics

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What are the contributors to edema refractory to loop diuretic therapy?

- Compensatory sodium reabsorption occurs after the drug wears off.
- Distal nephron hypertrophy and upregulation of sodium transporters in distal tubule enhances Na+ reabsorption.
- Volume contraction can lead to reduced renal perfusion, which decreases drug concentration at the site of action.
- Persistent activation of SNS and RAAS leads to aldosterone-mediated ↑ Na+ and water reabsorption.
- Decreased blood volume and increased plasma osmolality stimulates vasopressin-induced water reabsorption in collecting tubules, esp. in patients with heart failure, cirrhosis, or nephrotic syndrome.

Which patients are susceptible to loop diuretic-refractory edema?

- Patients with heart failure, chronic kidney disease, liver cirrhosis with ascites, nephrotic syndrome, secondary hyperaldosteronism.

What is the mechanism by which loop diuretics impair the kidney's ability to produce dilute urine during high intake of water and concentrated urine during hydropenia?

- Normally, TAL absorbs solutes without water (TAL is impermeable to water), which dilutes the tubular fluid.
- Blocking NKCC2 disables the diluting segment so it cannot maximally dilute urine even when water intake is high.
- Loop diuretics abolish the medullary osmotic gradient so water reabsorption is limited. Urine cannot be concentrated when water intake is low (hydropenia) even when vasopressin is present.

A 75-year-old male with hypertension and heart failure is taking lisinopril (ACE inhibitor) and furosemide. After self-treating chronic knee pain with naproxen, the patient develops dizziness and a sharp rise in serum creatinine.

Explain the mechanism of the drug-drug interaction leading to the adverse drug event.

Triple hit:

Blocking prostaglandin biosynthesis + blocking ANG II effects + ↓ effective circulating volume leads to collapse of glomerular filtration pressure and GFR → risk of acute kidney injury

1. NSAIDs block prostaglandin-mediated afferent dilation, leading to afferent constriction, decreased glomerular filtration pressure and GFR → risk of acute kidney injury.
2. ACE inhibitors/ARBs block angiotensin II-mediated efferent constriction, leading to efferent dilation, decreased glomerular filtration pressure and GFR → risk of acute kidney injury.
3. Diuretics cause ECV depletion, leading to reduced effective circulating volume, leading to decreased renal blood flow, decreased glomerular filtration pressure and GFR → risk of acute kidney injury.

EFC depletion activates RAAS but RAS inhibitors (ACE inhibitors/ARBs) block ANG II effects.